

Bollinger Mean Reversion Agent

Examine bands, RSI and trend context before treating an extreme as a reversal.

Release 2.1.1-source.20260928 | Updated October 3, 2026

Scope and limitations

This is downloadable source software. Hosting, trading funds and external services are separate. Build and signal checks are not certification for unattended real-money execution. Complete venue and recovery validation for your installation. Some packaged agent INSTALL/manifest files retain the older word “candidate”. This guide is tied to the registered September 28 archive by checksum; publication does not certify venue performance.

Markets can stay at an extreme. Do not remove trend checks merely to increase trading activity. Middle-band targets can move as new data arrives; runtime protection and replacement still require independent verification.

Requirements

A working Core/Trader dashboard in the manifest range $\geq 2.1.0$ $< 2.2.0$. The browser-only Core 2.0 template is incompatible.

The module manifest declares Node ≥ 20 ; the complete shipped dashboard requires Node 22.3 or newer.

Completed, time-ordered OHLCV candles with millisecond timestamps, finite positive prices and valid volume, supplied by your runtime.

A backup and an isolated paper/research environment. The strategy computes signals; the dashboard supplies market data, order handling and risk controls.

Purchase and download

1. Sign in to the account used at checkout. Open Account and locate the confirmed order. If payment is processing, wait for fulfillment rather than buying again.
2. Generate a fresh product Download link. Save the ZIP, receipt, release version and checksum. Expired links can be regenerated from the account.
3. Compare the archive SHA-256 with the release recorded below. Extract into a new directory; retain the original archive for rollback.
4. Open README.md, RUNBOOK.md or INSTALL.md, manifest.json where included, and LICENSE.md before changing an installation.

Archive SHA-256: dcdcd74fd55e4d4921314066dbe2743739e69679e31763b0b6b5ed3ada97432e

Installation

1. Check whether you already own the strategy: all six are included in Trader; Darvas is also included in Core.
2. Back up the database, settings and plugins. Pause new entries and reconcile positions before changing the installation.
3. Extract the module so manifest.json sits directly at plugins/mean-reversion-agent/manifest.json. Check plugins/mean-reversion-agent/dist/strategy.js exists; do not nest the product directory twice.
4. Restart the dashboard and inspect loader errors. Select strategy ID mean-reversion, a supported interval and the intended market. Start paused with no funds allocated.
5. Use the manifest defaults for the first comparison. Inspect decisions and then independently validate the runtime trade lifecycle before any unattended execution.

Run from the dashboard root

```
node -e "const p=require('./plugins/mean-reversion-agent/dist/strategy.js'); if(!p.strategy) throw Error('strategy export missing'); console.log('strategy export present')"
```

How the strategy works

The module combines Bollinger Bands, Keltner Channels and RSI, with checks for narrow ranges, band walking and double-bottom/top patterns. It proposes a single target at the middle or opposite band; it does not submit staged one-third exits.

Strategy ID: mean-reversion | Path: plugins/mean-reversion-agent

Supported intervals: 15m, 1h, 4h

Exact shipped defaults

Defaults are comparison inputs, not recommended trading parameters. Consult the source validation rules before changing them.

Setting	Default	Meaning
bbPeriod	20	Bollinger Band SMA period
bbStdDev	2	Standard deviation multiplier
kcPeriod	20	Keltner Channel EMA period
kcMultiplier	1.5	ATR multiplier for KC
rsiPeriod	14	RSI period
rsiOversold	30	RSI oversold threshold
rsiOverbought	70	RSI overbought threshold
bandwidthThreshold	5	Squeeze threshold %

Setting	Default	Meaning
walkingThreshold	3	Periods at band to consider trending
doubleBottomLookback	20	Periods to check for double bottom
minBodyPercent	0.3	Min candle body % for valid signal
takeProfitTarget	"middle"	TP at middle band or opposite band

Step-by-step practice tutorial

1. Verify the installed module

Follow the install steps above, then run the export check from the dashboard root. Expected result: strategy export present. This checks loading only; it does not submit an order.

2. Prepare comparable inputs

Select 15m, 1h, 4h as supported by the manifest. Supply one completed candle series for one symbol, retaining timestamps, OHLCV, source and network. Multiple listed intervals do not mean automatic multi-timeframe fetching.

3. Run the strategy-specific exercise

Use completed 1h candles, bbPeriod=20, bbStdDev=2 and takeProfitTarget="middle". Compare a ranging sample with a persistent move along a band. Record the RSI, band position and reason for the returned decision.

4. Interpret the result

A touch of the lower band is not automatically a buy. Read the regime and confirmation checks. bandwidthThreshold=5 is expressed in percentage points in the source calculation. Compare target choice only in a separate run.

5. Validate the runtime separately

Confirm ownership, sizing, fees, entry fill, stop/target, modifications, exit and realized P&L on testnet. Include restart, partial fill, rejected cancellation and lost-response reconciliation. A signal test is not an execution test.

Acceptance checklist

- [] Archive checksum and version match the guide.
- [] Install completes without suppressing errors; the module loads from its declared path.
- [] Correct owner, market, network and candle interval are recorded.
- [] Decision timestamp, direction, reasoning and parameters are captured; neutral is accepted.
- [] Settings and ownership persist after restart; backup restore is tested in isolation.

[] Before live execution: entry, fill, protection, modification, exit, fees and P&L reconcile; partial fills, cancellation failure and lost responses are tested.

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bandwidthThreshold=5 is expressed in percentage points in the source calculation. Compare target choice only in a separate run.

Maintain and customize

Keep a copy of the immutable ZIP, lockfile, private configuration and a database backup before an update. Store encrypted backups separately and prove a restore in an isolated compatible instance.

Pause new entries and reconcile existing venue orders and positions before maintenance. Pausing entries is not the same as closing positions. Removing a plugin does not close exchange positions.

Monitor the last successful cycle, data age, rejected orders, protection status, reconciliation gaps, database backups and disk space. Missing telemetry is an unknown state.

For rollback, stop the new runtime, reconcile the venue, restore a compatible application/schema pair in isolation and verify ownership and protection before resuming. A database restore cannot reverse a trade or transfer.

Customize readable source in a separate branch. Preserve the original defaults, rebuild any changed compiled module, and test the same inputs before and after. Follow LICENSE.md; source ownership does not imply unrestricted redistribution.

Troubleshooting

Payment confirmed but download missing

Check order status in Account and contact support with the order reference if fulfillment is delayed. Do not make another purchase to repair access.

Download link expired

Generate a fresh link from the purchasing account. The documentation downloads are public; the paid software archive remains entitlement-protected.

Strategy absent from the selector

Confirm manifest.json is directly inside the manifest installPath, the entryPoint exists, compatibility matches and the loader has no errors. Restart after installation; do not add another nesting level.

No trade despite available capital

Inspect the latest decision, candle history and freshness, symbol, interval, network, existing positions and risk rejection. Neutral is a valid result.

Build or database initialization fails

Keep the exact failed command and redacted diagnostic. Check Node, psql, Supabase settings and owner verification. Do not bypass types, authentication or schema integrity checks.

Balances disagree or an order outcome is unknown

Compare the same account and network at the venue, including fills, fees, margin and open orders. Reconcile before retrying; a lost response does not mean the order failed.

Help and supporting video lessons

<https://www.civalsystems.com/docs/products/mean-reversion-agent>

Supporting videos show the broader demo interface. The product-specific written tutorial matches this archive.

Contact <https://www.civalsystems.com/contact> with your order reference, product/version, OS, Node version and redacted error. Never send secrets. Refund requests: <https://www.civalsystems.com/refund-request>.

Source references

`mean-reversion-agent/INSTALL.md`

`mean-reversion-agent/LICENSE.md`

`mean-reversion-agent/README.md`

`mean-reversion-agent/dist/strategy.js`

`mean-reversion-agent/manifest.json`

`mean-reversion-agent/src/strategy.ts`